A DEEP LEARNING METHOD FOR DUAL-FREQUENCY RADAR SOUNDER DATA RETRIEVAL

Elena Donini
Fondazione Bruno Kessler
Trento, Italy
edonini@fbk.eu

Sonia Borsi
Fondazione Bruno Kessler
University of Trento
Trento, Italy
sborsi@fbk.eu

Lorenzo Bruzzone
University of Trento
Trento, Italy
lorenzo.bruzzone@unitn.it

Francesca Bovolo
Fondazione Bruno Kessler
Trento, Italy
bovolo@fbk.eu

***Abstract*—While the expansion of planetary radar sounder (RS) archives has driven the need for automated analysis, Content-Based Image Retrieval (CBIR) methods specifically designed for radargrams remain largely unexplored compared to other automated tasks. This paper proposes a deep-learning method for fast and accurate retrieval of dual-frequency RS data, enabling identification based on the semantic content of the most similar radargrams to a query that represents geologically significant subsurface reflections. We employ a Masked Autoencoder (MAE) to learn robust and semantically meaningful features of surface and subsurface reflections in a self-supervised manner, addressing the scarcity of labeled planetary data. The MAE projects radargrams into an embedding space, where retrieval is performed via cosine-similarity ranking. The proposed method is successfully validated on MARSIS radargrams spanning diverse geological targets. The results demonstrate strong generalization: when queried with North Polar icy stratigraphy, it effectively retrieves analogous features from the South Pole and equatorial ice-rich region.**

***Index Terms*—Content-based retrieval, masked auto encoders, Mars subsurface, radar sounder.**

I. Introduction

Radar sounders (RSs) are essential for planetary exploration as they reveal hidden geological structures beneath planetary surfaces, enabling discoveries, such as evidence for the characterization of the ice-rich layered deposit in the Medusae Fossae Formation (MFF) [1]. Over the last few decades, orbital missions to the Moon and Mars have acquired vast RS archives and RSs onboard JUICE [2] and Clipper [3] missions to the Jovian icy moons and the future EnVision mission [4] to Venus will further increase the amount of available data. The growth poses a critical challenge, as it far outpaces the capacity of the scientific community to manually analyze the acquired data. To address this, CBIR methods are important for automating the analysis of RS archives, enabling fast, accurate retrieval of the most similar radargrams to a query with geologically interesting subsurface reflections. Deep Learning (DL) has emerged as an approach to automate the extraction of meaningful features from RS data [5, 6]. However, the literature lacks a CBIR method tailored to radargrams.

In most CBIR methods, retrieval is performed by first extracting descriptive features from every database entry and comparing them with the query features using a similarity metric. Recently, feature extraction has been driven by deep learning, with self-supervised approaches gaining particular interest as they do not require labeled training data. This is especially valuable in the RS domain, where annotated data are scarce. Early self-supervised CBIR methods relied on contrastive learning to learn robust features [7, 8]. More recently, masked autoencoders (MAEs) [9] were applied to CBIR in hyperspectral or multispectral imagery [10, 11]. MAE operates by randomly masking a large portion of the input and training an encoder–decoder architecture to reconstruct the masked content from the unmasked portions, thereby learning informative latent features. However, none of the approaches in the literature address the specific challenges of RS data.

This paper proposes a novel DL method for content-based retrieval of dual-frequency RS data with self-supervised learning. We employ MAE to learn semantic features from radargrams. By training the model to reconstruct masked parts of the signal, the encoder learns to capture the intrinsic statistical and structural properties of the surface and subsurface reflections. The pre-trained MAE encoder extracts features from RS data in the archive to build a feature database for retrieval, given a query. We evaluated the proposed method on a dataset of MARSIS data spanning the North Polar Layered Deposits (NPLD), the South Polar Layered Deposits (SPLD), and the Medusae Fossae Formation (MFF) on Mars.

This paper's structure is as follows. Section II details the proposed method, including the MAE architecture and CBIR system. Section III describes the MARSIS dataset and presents the experimental results. Finally, Section IV concludes the paper and outlines future research.

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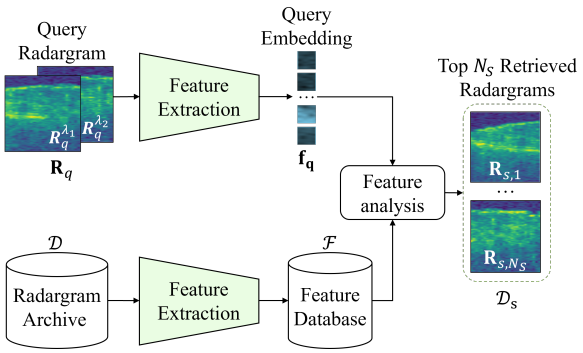


Fig. 1. Flowchart of the proposed method for radargram CBIR.

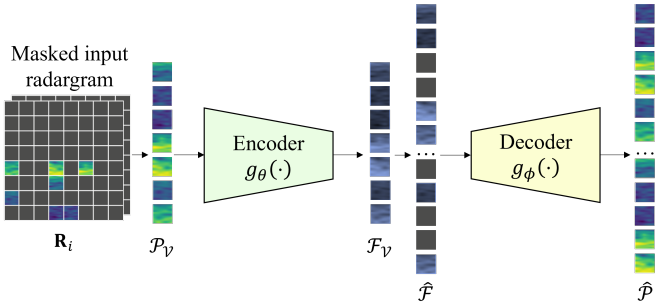


Fig. 2. Flowchart of the MAE training strategy.

II. Methodology for RS content-based retrieval

This work proposes a novel CBIR method tailored to RS dual-frequency data. The retrieval task is to identify radargrams in an archive that exhibit high semantic similarity to a given query, accounting for the structure of surface and subsurface reflections.

Let $\mathcal{D} = \{\mathbf{R}_i\}_{i=1}^{N_R}$ denote the database comprising N_R radargrams in $\mathbb{R}^{N_D \times N_D \times C}$, where N_D is the dimension in the range and azimuth directions, respectively, and C is the dimension in the frequency domain. In currently existing radar sounders, C typically equals 1 or 2. Here, we assume $C = 2$, which corresponds to the dual-frequency channels that are centered at wavelengths λ_1 and λ_2. Given a query $\mathbf{R}_q$, we aim to retrieve a subset $\mathcal{D}_s = \{\mathbf{R}_{s,l}\}_{l=1}^{N_S} \subset \mathcal{D}$, containing the top-$N_S$ most semantically similar radargrams to $\mathbf{R}_q$. The proposed method consists of three steps (see Fig. 1). First, we train the MAE to learn semantically meaningful features: $\mathbf{R}_i, i \in [1, N_R]$ are partitioned into a set of patches $\mathcal{P}$. During training, we randomly mask a significant portion of the patches $\mathcal{P}_M$, so that the network reconstructs the masked patches from the observed context. This pre-text task forces the encoder-decoder network to learn the underlying statistical distribution and spatial dependencies of surface and subsurface reflections without requiring manual annotation. Second, we discard the decoder and use the MAE encoder as a feature extractor. $\mathbf{R}_q$ and $\mathbf{R}_i \in \mathcal{D}$ are mapped into a compact latent embedding feature space. This generates a feature database $\mathcal{F} = \{\mathbf{f}_i \in \mathbb{R}^{N_F}\}_{i=1}^{N_R}$ and a query embedding $\mathbf{f}_q \in \mathbb{R}^{N_F}$, where N_F is the dimension of the embedding vector. Third, we perform the retrieval by computing the cosine similarity between $\mathbf{f}_q$ and every $\mathbf{f}_i \in \mathcal{F}$. We rank the resulting samples in descending order of similarity, and return the top-N_S samples as the most similar radargrams.

A. Pre-Training of the Masked Autoencoder

We adapt an MAE architecture to process dual-frequency radar data, treating the radargrams at λ_1 and λ_2 as two distinct input channels. Incorporating dual-frequency data considers the frequency-dependent nature of radar backscattering, which is governed by the relationship between frequency and surface roughness [12]. Therefore, jointly processing these frequencies enables the network to discriminate between different types of subsurface reflections.

Each $\mathbf{R}_i$ is first divided into N_P non-overlapping square patches $\mathbf{P}_k \in \mathbb{R}^{P^2 C}$ of width and height P that are flattened into $\mathcal{P}$. $\mathcal{P}$ is divided into two disjoint subsets: a masked set $\mathcal{P}_M = \{\mathbf{P}_m\}_{m=1}^{N_M}$ and a unmasked set $\mathcal{P}_V = \{\mathbf{P}_v\}_{v=1}^{N_V}$, where $N_P = N_M + N_V$ and $N_M >> N_V$ for a high masking ratio.

We consider a network made of an encoder $g_\theta(\cdot)$ and a decoder $g_\phi(\cdot)$. $g_\theta(\cdot)$ is a Vision Transformer (ViT) [13] that encodes patch positions in $\mathbf{R}_i$ via positional embeddings (PEs), enabling the network to learn spatial relationships during pre-training. This inherently aligns with the structural nature of RS data, where geological targets typically exhibit strong lateral continuity along the azimuth and follow a precise stratigraphic organization along the range [5]. $g_\theta(\cdot)$ maps $\mathcal{P}_V$ into the latent feature space $\mathcal{F}_V = \{\mathbf{f}_{k,v}\}$, resulting in the feature embeddings:

$$\mathbf{f}_{k,v} = g_\theta(\mathbf{P}_{k,v}) + \mathbf{e}^{\text{pos}}_{k,v}, \tag{1}$$

where $\mathbf{e}^{\text{pos}}_k \in \mathbb{R}^{N_F}$ is the PE added to retain the location of $\mathbf{P}_k$ in $\mathbf{R}_i$.

$g_\phi(\cdot)$ reconstructs $\mathbf{P}_m$, conditioned on $\mathbf{f}_{k,v}$, $\mathbf{e}^{\text{pos}}_{k,v}$, and the masked positional embedding $\mathbf{e}^{\text{pos}}_{k,m}$. $\mathbf{e}^{\text{pos}}_{k,v}$ is combined in sequence with a shared learnable mask token $\mathbf{m}_k \in \mathbb{R}^{N_F}$ inserted at every masked index $m, \in [1, N_M]$. $g_\phi(\cdot)$ processes the whole sequence to generate the reconstruction masked patch set $\hat{\mathcal{P}_m} = g_\phi(\mathbf{f}_{k,v}, \mathbf{e}^{\text{pos}}_{k,v}, \mathbf{e}^{\text{pos}}_{k,m}) = \{\hat{\mathbf{P}}_m\}_{m=1}^{N_M}$, where each reconstructed patch $\hat{\mathbf{P}}_m \in \mathbb{R}^{P^2 C}$ represents the predicted intensity values for the masked m-th patch. Consequently, the training loss minimizes the Mean Squared Error (MSE) computed between $\mathbf{P}_m$ and the corresponding predictions $\hat{\mathbf{P}}_m$:

$$\mathcal{L}_{\text{MAE}} = \frac{1}{N_M} \sum_{k=1}^{N_M} \left\| \hat{\mathbf{P}}_m - \mathbf{P}_m \right\|_2^2. \tag{2}$$

B. Feature Extraction and Retrieval

In the inference phase, the decoder is discarded, and the pre-trained $g_\theta(\cdot)$ processes $\mathbf{R}_i$ to compute the corresponding feature embedding $\mathbf{f}_i$, which represents the signature of its semantic content. The model ensures that radargrams with analogous geological structures produce similar embeddings,

while assigning distinct signatures to diverse subsurface structures. The features of $\mathbf{R}_i \in \mathcal{D}$ are combined to generate the feature database $\mathcal{F}$.

For any given query $\mathbf{R}_q$, the feature vector $\mathbf{f}_q$ is computed with the pre-trained MAE encoder. The retrieval task involves identifying the $\mathbf{R}_i \in \mathcal{D}$, whose embeddings in $\mathcal{F}$ are the most similar to $\mathbf{f}_q$. The similarity between $\mathbf{f}_q$ and $\mathbf{f}_i \in \mathcal{F}$ is measured using the cosine similarity [14], which measures the angular distance between vectors in the feature space and can be formally defined as:

$$S_C(\mathbf{f}_q, \mathbf{f}_i) = \frac{\mathbf{f}_q \cdot \mathbf{f}_i}{\|\mathbf{f}_q\|_2 \|\mathbf{f}_i\|_2}. \quad (3)$$

However, any other distance metric can be used. For each $\mathbf{R}_q$, we compute the similarity with all the embeddings in $\mathcal{F}$ and then rank them in descending order. Finally, we retrieve the top-N_S $\mathbf{R}_i$.

III. Dataset description and Experimental results

A. Dataset and Experimental Setup

To evaluate the effectiveness of the proposed method, we consider a dataset of 45 radargrams acquired by MARSIS [15] over the SPLD and NPLD and MFF, which exhibit icy specular reflections and rocky targets. MARSIS operates by transmitting chirp pulses with a 1 MHz bandwidth, centered at carrier frequencies of 1.8, 3, 4, or 5 MHz and alternating between two frequencies. RS data are processed with ionosphere compensation [16], Hann windowing to suppress range sidelobes, unfocused processing, multilooking, and resampling, depending on the spacecraft altitude and velocity relative to the surface under observation [17]. The resulting power radargram is converted to decibels to highlight the reflections.

For the pre-training phase of MAE, we consider 37 radargrams over the SPLD, divided into 33,280 radargram patches of dimension $C = 2$ and $N_D = 256$. The training radargrams are selected to capture the properties of ice-rich subsurface regions and arid areas with high roughness values. 8 radargrams intersecting specific geological targets of interest (i.e., SPLD, NPLD, and MFF) are divided into 3,950 radargram patches to generate the validation dataset. To validate the proposed method, we generated a labeled dataset using expert annotations and literature references [1]. We labeled the radargram patches sets considering five classes, as illustrated in Fig. 3: i) ω_1 indicates patches imaging the surface, shallow subsurface, and background (free space above the surface); ii) ω_2 indicates patches imaging only background, iii) ω_3 indicates patches imaging the surface and background, iv) ω_4 surface and shallow subsurface, and v) ω_5 noisy area below the shallow subsurface. To consider different subsurface terrain, we manually annotated the classes imaging the shallow subsurface $\omega_c, c = [1, 4]$ into three subclasses: ω_c^{ice} ice for patches imaging NPLD and SPLD; $\omega_c^{\text{ice, rock}}$ for patches imaging ice layers in rocky areas (e.g., the MFF [1]); and ω_c^{rock} for patches imaging rocky regions.

We trained the ViT-based architecture with 5.6M parameters on a Tesla V100 GPU for 300 epochs with 10 warm-up epochs for about 7 hours. We set $P = 32$ and masked 90% of the MAE patches. The learning rate was dynamically scaled linearly with respect to the batch size, and a scheduler was used to ensure smooth convergence [9]. We used the AdamW optimizer with $\beta_1 = 0.9$ and $\beta_2 = 0.95$ and a weight decay of 0.05.

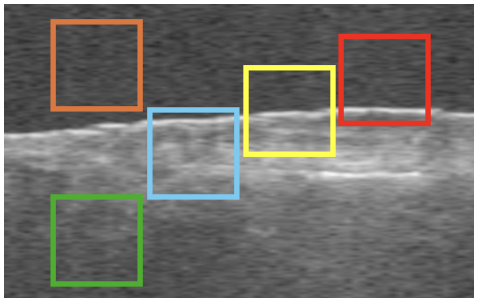


Fig. 3. Radargram imaging the SPLD with label ω_1 in yellow, ω_2 in orange, ω_3 in red, ω_4 in blue, ω_5 in green.

TABLE I
MAP CALCULATED ON THE TOP-$N_S = [5, 10, 20]$ RETRIEVED PATCHES.

Class	mAP@5	mAP@10	mAP@20
ω_1	97.07%	96.33%	95.47%
ω_2	98.82%	98.62%	98.35%
ω_3	95.58%	94.92%	93.94%
ω_4	94.27%	92.98%	91.42%
ω_5	96.99%	96.22%	95.71%

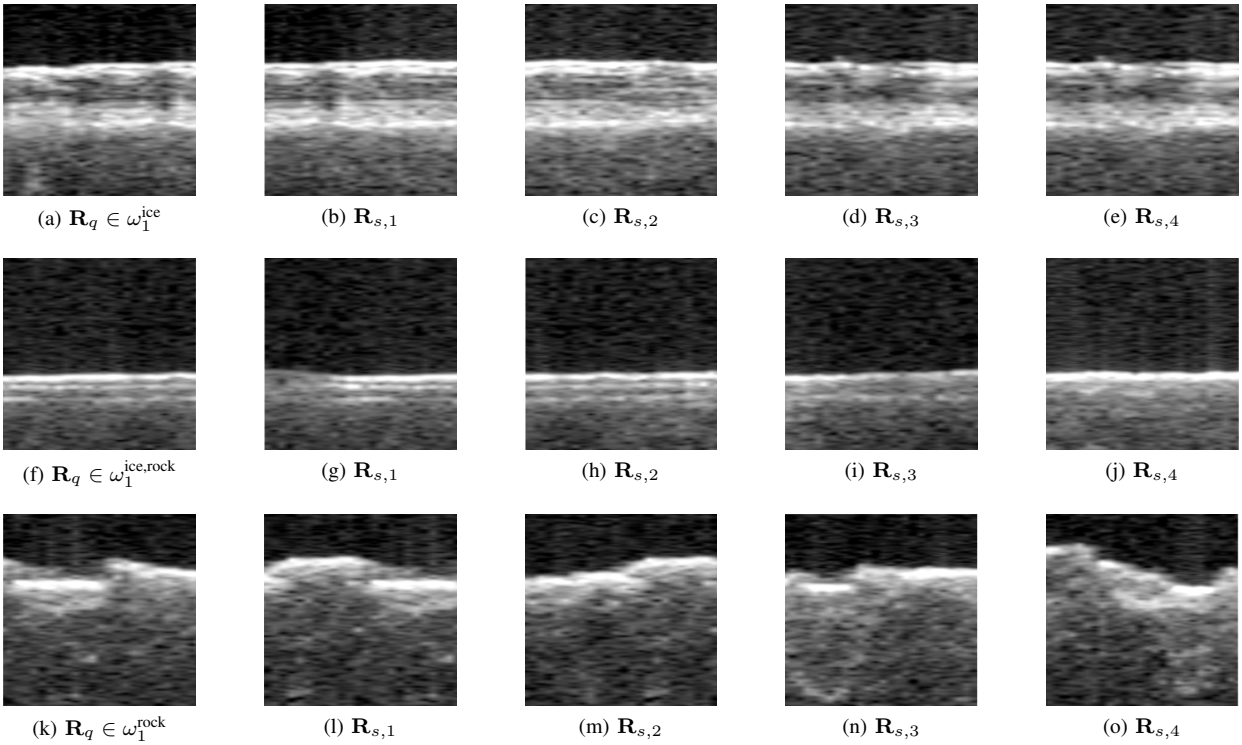
We evaluate the retrieval performance of the proposed architecture using the Mean Average Precision (mAP). mAP accounts for both the correctness of the retrieved items and their position within the ranked list for N_Q queries and is defined as:

$$\text{mAP} = \frac{1}{N_Q} \sum_{q=1}^{N_Q} \frac{1}{N_S} \sum_{l=1}^{N_S} Pr(\omega_c^q, \omega_c^l)\ \text{rel}(\omega_c^q, \omega_c^l), \quad (4)$$

where $Pr(\omega_c^q, \omega_c^l)$ is the precision of the q-th query at rank l, and $\text{rel}(\omega_c^q, \omega_c^l)$ is an indicator function that equals 1 if the radargram retrieved at rank l is relevant to query q, and 0 otherwise. This formulation favors small values of l as highly relevant radargrams should appear at the top of the retrieved list. To better characterize performance, we compute mAP for $N_S \in [5, 10, 20]$ because only a limited number of samples represent certain geological classes (e.g., ice-bearing patches within the MFF). Varying N_S is important to ensure a fair evaluation across both rare and common classes.

B. Experimental results

Table I reports accurate retrieval, indicating that the network learns meaningful features of the Martian surface and subsurface. Notably, the mAP is above 90% for all the classes for $N_S \in [5, 10, 20]$. Table II reports the mAP for classes ω_1 and ω_4 across different shallow subsurface terrains and



(a) $\mathbf{R}_q \in \omega_1^{\text{ice}}$ (b) $\mathbf{R}_{s,1}$ (c) $\mathbf{R}_{s,2}$ (d) $\mathbf{R}_{s,3}$ (e) $\mathbf{R}_{s,4}$

(f) $\mathbf{R}_q \in \omega_1^{\text{ice,rock}}$ (g) $\mathbf{R}_{s,1}$ (h) $\mathbf{R}_{s,2}$ (i) $\mathbf{R}_{s,3}$ (j) $\mathbf{R}_{s,4}$

(k) $\mathbf{R}_q \in \omega_1^{\text{rock}}$ (l) $\mathbf{R}_{s,1}$ (m) $\mathbf{R}_{s,2}$ (n) $\mathbf{R}_{s,3}$ (o) $\mathbf{R}_{s,4}$

Fig. 4. Examples of $\mathbf{R}_q$ (column 1) and $\mathbf{R}_s$ (columns 2 to 4). For space constraints, we show only one frequency per example.

TABLE II
MAP CALCULATED FOR ω_1 AND ω_4 IN ICY AND ARID TERRAINS ON THE TOP-$N_S = [5, 10, 20]$ RETRIEVED PATCHES.

Class	mAP@5	mAP@10	mAP@20
ω_1^{ice}	75.43%	72.23%	66.93%
ω_4^{ice}	74.77%	71.03%	66.12%
ω_1^{rocky}	93.71%	90.96%	88.05%
ω_4^{rocky}	88.08%	85.63%	82.61%

for various values of N_S. In the case of the icy terrain and $N_S = 5$, mAP is 75.43% for ω_1^{ice} and 74.77% for ω_4^{ice}. In the case of the rocky terrain and $N_S = 5$, mAP is 93.71% for ω_1^{rock} and 88.08% for ω_4^{rock}. Overall, the results show that the proposed method effectively learns discriminative features and generalizes well across icy and rocky Martian subsurface terrains, with retrieval performance generally higher for rocky areas than for icy regions. This performance difference may be partially attributed to the unbalanced training dataset, in which rocky terrain samples are more numerous and diverse, leading to an embedding space better optimized for rocky subsurface environments.

Fig. 4 illustrates examples of queries and their corresponding retrieved radargrams for different subsurface terrains: from top to bottom, ice from the NPLD ω_1^{ice}, ice in the MFF $\omega_1^{\text{ice, rock}}$, and a crater in a rocky region ω_1^{rock}. The qualitative results indicate that the proposed method consistently retrieves patches that are highly similar to the query across all terrains. Notably, for the ice-related examples, the proposed method successfully retrieves icy radargrams from the NPLD (first row) and MFF (second row) despite the training set containing only ice signatures from the SPLD. This indicates that the network captures transferable, physically meaningful ice-related features rather than region-specific patterns.

IV. Conclusions and Future Works

This paper presents a novel deep learning framework for CBIR of dual-frequency radargrams. The proposed method exploits an MAE to learn semantically meaningful features from an RS archive in a self-supervised manner. Experimental results on MARSIS data demonstrate that the approach effectively encodes the radiometric and structural properties of surface and subsurface reflections. The method achieves high retrieval performance across multiple semantic classes and demonstrates robustness in retrieving shallow subsurface targets across diverse terrains.

Future work will extend the proposed CBIR method to different geological targets, such as Non-Reflective Areas (NRAs) on Mars. Furthermore, the method will be adapted to higher-resolution datasets, including SHARAD radargrams, to evaluate its performance in retrieving fine-scale features and cross-sensor retrieval capabilities.

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